

Lab 4 – VibeCoding

Chat History

Project: Pygame Endless Runner

Relevant ChatGPT interaction used for Tasks 1–4 of the Lab 4 assignment.

Task 1 – Fix High-Speed Collision

User: Fix the broken collision code where obstacles move before collision is checked. The obstacle speed continuously increases, which can cause an obstacle to pass completely through the player's hitbox between frames.

Assistant: Identified the problem as collision tunneling. Kept Player and Obstacle unchanged, saved each obstacle's previous x-position before movement, retained the normal colliderect check, and added swept horizontal collision detection between the previous and current positions. Also added a maximum speed so speed cannot grow without bound.

Task 2 – Game Over Screen

User: Add a proper Game Over screen showing GAME OVER and the final score, stop gameplay, wait for input, preserve Task 1, and do not implement replay yet.

Assistant: Kept the existing game_over state and update() early return. Updated handle_event() to wait for input after Game Over and updated render() to display GAME OVER, final score, and an exit instruction.

Task 3 – Replay / Difficulty Selection

User: After Game Over, show Easy, Medium, Hard and Exit. Allow keyboard selection, make each difficulty actually change the starting difficulty, completely reset the game on replay, preserve Tasks 1 and 2, and do not add sound.

Assistant: Added difficulty settings and reset_game(). Easy starts at speed 4 with spawn interval 90 and max speed 10. Medium starts at speed 6 with spawn interval 70 and max speed 12. Hard starts at speed 8 with spawn interval 50 and max speed 14. Keys 1, 2, 3 select the difficulties and 4 exits. reset_game() recreates the player, clears obstacles, resets score, timers, speed variables, distance and game_over.

Task 4 – Sound Feedback

User: Add sound for jumping, successfully passing an obstacle and gaining a point, and Game Over. Use Pygame sound functionality, add sound assets, preserve Tasks 1–3, and do not add unnecessary changes.

Assistant: Added jump.wav, score.wav and game_over.wav under assets/sounds. Loaded them using pygame.mixer.Sound(), added a play_sound() helper, played jump_sound after player.jump(), score_sound after score increases, and game_over_sound at both normal and swept collision paths. Missing audio assets are handled by disabling sound instead of crashing.

The user tested the game and confirmed all three sounds work.

Git / Testing

Task 1 commit: Fix high speed obstacle collision

Task 2 commit: Add game over screen

Task 3 commit: Add replay difficulty selection

Task 4 commit: Add sound feedback

The user tested the completed features. A push to the assigned SETAPESU26 repository was rejected because the authenticated GitHub account did not have permission. The commits were successfully created locally. The lab requires the final work to be pushed to the student's personal repository, with no PR to SETAPESU26.

Submission Status

Completed coding/features:

Task 1 – High-speed collision fix

Task 2 – Game Over screen

Task 3 – Replay/difficulty selection

Task 4 – Sound feedback

Remaining submission items discussed:

before and after 10-second videos, updated code under Lab-4, this chat-history PDF, and pushing the final work to the personal repository.